No	Author & Year	Research Topic	Variables Used	Algorithm / Method	Key Findings	Limitations	Research Gap
1	Du et al., 2017	Anomaly detection and diagnosis from system logs	Log-event sequences, execution paths	DeepLog; LSTM next-event prediction	Detects anomalous log sequences and supports diagnosis from system logs.	Primarily evaluated on system logs; limited semantic understanding of raw text.	Need for richer semantic/context-aware detection.
2	Meng et al., 2019	Sequential and quantitative log anomaly detection	Event sequences, event counts	LogAnomaly; LSTM + quantitative features	Combines sequential and quantitative information to detect anomalies.	Depends on parsed log templates and historical normal behavior.	More robust raw-log and domain-aware approaches are needed.
3	Zhang et al., 2019	Anomaly detection on unstable log data	Log-event sequences, semantic vectors	LogRobust; attention-based Bi-LSTM	Improves robustness when log events and templates evolve.	Requires log parsing and focuses on anomaly detection rather than remediation.	Detection should be connected to actionable diagnosis/fixes.
4	Guo et al., 2021	Transformer-based log anomaly detection	Normal log sequences, event tokens	LogBERT; BERT + self-supervised learning	Learns normal log patterns and detects deviations without extensive labels.	Training/inference can be resource intensive; no remediation component.	Need lightweight, operationally useful detection with downstream actions.
5	Zhu et al., 2021	Log anomaly detection without log parsing	Raw log messages, contextual semantics	NeuralLog; semantic vectors + Transformer	Avoids errors introduced by conventional log parsing and achieves strong anomaly detection.	Focuses on detection; limited pipeline-specific context.	Combine semantic detection with pipeline context and remediation.
6	Nedelkoski et al., 2020	Self-attentive anomaly detection in unstructured logs	Log-event sequences	Self-attention / Transformer classifier	Uses attention to capture important event dependencies for anomaly detection.	Primarily evaluates anomaly classification rather than operational recovery.	Bridge anomaly detection and operational response.
7	Farzad & Gulliver, 2020	Unsupervised log message anomaly detection	Log messages, learned representations	Autoencoder-based anomaly detection	Shows that unsupervised representation learning can detect anomalous messages.	Limited contextual and semantic reasoning; no remediation.	More explainable and actionable anomaly handling is needed.
8	Chen et al., 2021	Deep learning-based system log analysis	System logs, event sequences	Deep learning models including LSTM-based approaches	Reports effectiveness of deep learning for automated log anomaly detection.	Highlights dataset and generalization challenges across systems.	Need cross-domain and cost-aware deployment studies.
9	Yu et al., 2016	Workflow monitoring of cloud infrastructures	Interleaved logs, workflow states	CloudSeer; workflow automata	Detects workflow divergence online and provides contextual logs for diagnosis.	Rule/workflow modeling is required; no ML/LLM remediation.	Automated semantic diagnosis and remediation can extend workflow monitoring.
10	Fu et al., 2009	Execution anomaly detection in distributed systems	Unstructured logs, execution sequences	LKE / log sequence analysis	Shows that mining unstructured logs can reveal execution anomalies.	Older log formats and limited semantic understanding.	Modern ML/LLM methods can improve adaptability and explanations.
11	He et al., 2016	System log analysis for anomaly detection	Log events, system behavior	Log mining / statistical and clustering methods	Demonstrates practical challenges in system log anomaly detection.	Manual preprocessing and system-specific assumptions.	Need automated and adaptable pipeline-oriented approaches.
12	Astekin et al., 2018	Distributed ML for anomaly detection in large-scale logs	Large-scale system logs, event features	Distributed machine learning algorithms	Evaluates scalability of ML-based anomaly detection on large log datasets.	Focuses on detection and compute scalability rather than remediation.	Evaluate accuracy together with operational cost and downstream actions.
13	Liu et al., 2018	Anomaly detection from massive system logs	Log events, statistical features	Integrated anomaly detection framework	Integrates multiple log-analysis stages for massive-scale anomaly detection.	Limited semantic explanation and automated fixing.	Integrate detection with intelligent explanation and recommended fixes.
14	Huang et al., 2018	Failure detection through system observability	Runtime observations, logs, system signals	In-situ observability and failure detection	Shows that richer observability can improve failure detection.	Requires instrumentation and focuses on detection.	Combine observability signals with lightweight AI diagnosis.
15	Le & Zhang, 2020	Generative anomaly detection for logs	Log-event sequences	LogGAN; generative adversarial network	Uses generative modeling to detect anomalous log patterns.	Training complexity and limited interpretability.	Need interpretable and cost-efficient methods for operations.
16	Lu et al., 2018	Anomaly detection in big-data system logs	Log messages, token features	Convolutional Neural Network	Demonstrates CNN-based detection for big-data system logs.	Requires labeled/structured data and focuses on anomaly classification.	Need semantic explanation and remediation assistance.
17	He et al., 2017	Online log parsing for streaming systems	Raw log messages	Drain; fixed-depth parse tree	Fast online parsing with strong accuracy across large datasets.	Parsing only; does not identify root causes or propose fixes.	Use efficient parsing as a preprocessing layer for intelligent diagnosis.
18	Du & Li, 2019	Online streaming parsing of unstructured logs	Streaming log messages	Spell; longest common subsequence	Efficiently extracts log patterns in streaming environments.	Template extraction does not provide semantic diagnosis.	Connect streaming parsing with ML/LLM-based failure handling.
19	He et al., 2018	Automated log parsing for large-scale analysis	Raw logs, message templates	Automated parsing and clustering	Provides methods and evaluation for large-scale automated log parsing.	Focuses on structuring logs rather than downstream failure response.	Need end-to-end pipeline failure management after parsing.
20	Zhu et al., 2019	Benchmarks and tools for automated log parsing	Multiple log datasets and parser outputs	LogParser toolkit and benchmark	Provides reproducible comparison of automated log parsers.	Benchmark mainly targets parsing accuracy, not incident resolution.	Develop benchmarks extending from parsing to diagnosis/remediation.
21	Ding et al., 2020	Efficient log parsing using n-gram dictionaries	Raw log messages, token n-grams	Logram; n-gram dictionaries	Improves parsing efficiency and accuracy using n-gram information.	Dependent on parsing/template extraction and not designed for fixes.	Efficient preprocessing should feed actionable AI systems.
22	Hamooni et al., 2016	Fast pattern recognition for log analytics	Log messages, recurring patterns	LogMine; pattern mining	Efficiently mines patterns from large log collections.	Pattern mining alone does not determine operational causes or actions.	Move from pattern discovery toward diagnosis and remediation.
23	Vaarandi & Pihelgas, 2015	Event-log clustering and pattern mining	Event logs, message patterns	LogCluster; clustering and pattern mining	Identifies frequent patterns and unusual events efficiently.	Limited semantic reasoning and no automated repair.	Combine pattern mining with intelligent recommendations.
24	Xin et al., 2023	Fine-grained root-cause localization in microservices	Service metrics, resource metrics	CausalRCA; causal graph learning + PageRank	Automates coarse- and fine-grained root-cause localization and improves localization accuracy.	Primarily metric-based and microservice-focused; not ETL-specific.	Adapt RCA to data pipelines and combine with log/LLM reasoning.
25	Ahmed et al., 2023	Root-cause and mitigation recommendations for cloud incidents	Incident descriptions, historical incident data	LLMs; zero-shot, fine-tuned and multi-task settings	Large-scale study shows LLMs can assist incident RCA and mitigation recommendations.	Cloud incidents are broader than ETL failures; cost and controlled pipeline experiments are not the central focus.	Evaluate a focused ETL setting with explicit cost constraints.
26	De la Cruz Cabello et al., 2025	AIOps log anomaly detection in the LLM era	AIOps logs, anomaly labels	Systematic literature review; LLMs and RAG	Finds prompt engineering and RAG promising for accuracy and interpretability.	Identifies broad AIOps gaps rather than ETL-specific remediation.	Need domain-specific, cost-aware ETL failure workflows.
27	Ma et al., 2026	LLM-based log analysis	Logs, anomaly/failure/RCA tasks	Systematic review of LLM4Log methods	Reviews 145 papers across log generation, parsing, anomaly detection, failure prediction, RCA and summarization.	Highlights unresolved issues including cost, latency, hallucination, drift and reproducibility.	Cost-aware hybrid ML+LLM pipeline research is a relevant direction.
28	De la Cruz Cabello et al., 2026	LLM-based log anomaly detection in AIOps	Linux syslogs, log sequences	LogBERT; self-supervised masked-token prediction	Reports strong anomaly detection with a favorable accuracy/latency trade-off using local inference.	Focused on Linux syslogs and anomaly detection; no ETL remediation.	Study pipeline-specific logs and connect detection to actionable fixes.
29	Bâra et al., 2026	Self-healing autonomous ML pipeline generation	ML tasks, pipeline execution history, runtime errors	Multi-agent AI + RAG + LLM error interpretation + self-healing	Integrates pipeline generation, recommendation and LLM-based self-healing; reports improved pipeline success and reduced development time.	Broader ML pipeline generation problem; more complex architecture than a college-scale ETL study.	A smaller, cost-aware ETL failure detection/remediation study is a feasible incremental direction.
30	Guan et al., 2026	LLM-assisted software bug diagnosis using resolved incident history	Resolved issues, symptoms, root causes, fixes	Operational Memory RAG (OM-RAG) + embeddings + LLM	Structured historical failure-resolution records improve diagnosis and fix correctness.	Focused on software bugs/Dataverse rather than ETL pipelines; not centered on ML pre-filtering or cost.	Investigate whether ML pre-filtering plus targeted LLM context can reduce cost while retaining fix quality.